PAVITHRA K M

AI & Generative AI Enthusiast | Machine Learning | Data Science

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OBJECTIVE

M.Sc. Data Analytics and Computational Science student with a mathematics foundation and hands-on experience in machine learning, deep learning, and Large Language Models — including backend development, data preparation, fine-tuning, and evaluation of an LLM-based question-answering chatbot. Proficient in Python, SQL, and Google Cloud, with five end-to-end projects spanning classification, forecasting, and generative AI. Seeking an **internship or entry-level role** in Data Science, Machine Learning, or AI to build and evaluate real-world data-driven applications.

TECHNICAL SKILLS

- **Programming & Data:** Python, SQL, Data Cleaning, Exploratory Data Analysis (EDA), Statistical Analysis, Data Visualization
- **Machine Learning:** Classification, Regression, Deep Learning, Feature Engineering, Model Evaluation, Predictive Modeling
- **Generative AI & LLMs:** Large Language Models, Model Fine-tuning, Question-Answering Systems, Chatbot Evaluation
- **Tools & Platforms:** Google Cloud Platform (BigQuery, Looker Studio), Power BI, GitHub, Jupyter Notebook, Streamlit
- **Core Concepts:** Data Structures & Algorithms, Problem Solving, Research & Analysis

PROJECTS

LLM-Based Question Answering Chatbot

- Contributed to backend development, data preparation, model fine-tuning, and chatbot evaluation for a question-answering system built on Large Language Models.

Tech: Generative AI, Large Language Models, Model Fine-tuning, Chatbot Evaluation

Taxi Demand Prediction

- Developed a machine learning approach to analyze taxi demand patterns and predict demand from historical data.
- Covered data preprocessing, exploratory data analysis, feature engineering, and predictive modeling.

Tech: Machine Learning, Demand Prediction, Data Preprocessing, Feature Engineering

Learning Unidirectional and Bidirectional Coupling of Discrete Maps

- Researching Echo State Networks' (ESNs) ability to learn and predict coupled chaotic system dynamics across five discrete maps (Hénon, Lorenz, Mira, Nordmark, Lozi).
- Evaluating how reservoir dimensionality and topology affect prediction accuracy across unidirectionally and bidirectionally coupled systems.

Chess Piece Classification Using Machine Learning

- Built a 12-class chess piece image classifier using HOG features and PCA; benchmarked SVM, Random Forest, and Logistic Regression, and deployed the best pipeline as an interactive Streamlit app for real-time inference.

Handwritten Digit Recognition Using Machine Learning

- Engineered HOG and pixel-intensity feature sets and benchmarked SVM, KNN, Logistic Regression, and Random Forest to identify the most effective approach and improve classification robustness.

EDUCATION

M.Sc. Data Analytics and Computational Science — 2025 – 2027 (Expected)

Kerala University of Digital Sciences, Innovation and Technology (Digital University Kerala)

B.Sc. Mathematics — 2022 – 2025

Mar Dionysius College, Pazhanji, University of Calicut

CERTIFICATIONS

- Google Cloud Career Launchpad – Data Analytics Track, Google Cloud (2026)
- IBM Introduction to Git and GitHub
- NCC 'A', 'B', 'C' Certificates

LEADERSHIP & ACTIVITIES

- National Cadet Corps (NCC): Held a structured leadership role coordinating training and community engagement activities within a cadet unit, developing discipline, teamwork, and organizational skills over multiple years of service.

SOFT SKILLS & LANGUAGES

Analytical Thinking • Problem Solving • Team Collaboration • Communication • Time Management • Quick Learning
Languages: Malayalam, English